

Homework 2

Connor Mooneyhan

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1. Let $\mathbb{P}^2$ be the projective plane. A line is defined to be the zero set of

$$\alpha X + \beta Y + \gamma Z = 0.$$

- (a) Show directly from the definition that two distinct points in $\mathbb{P}^2$ are contained in a unique line.
- (b) Similarly, show that any two distinct lines in $\mathbb{P}^2$ intersect in a unique point.

2. For each of the pairs below, find all of the points of intersection between the curves C_1 and C_2 in $\mathbb{P}^2(\mathbb{C})$. (So include complex points and points at infinity.)

- (a) $C_1 : x - y = 0$, $C_2 : x^2 - y = 0$.
- (b) $C_1 : x - y - 1 = 0$, $C_2 : x^2 - y^2 + 2 = 0$.
- (c) $C_1 : x - y - 1 = 0$, $C_2 : x^2 - 2y^2 - 5 = 0$.
- (d) $C_1 : x - 2 = 0$, $C_2 : y^2 - x^3 + 2x = 0$.

Proof.

- (a) The points on C_1 can be written as $\langle x : x : 1 \rangle$ or $\langle 1 : 1 : 0 \rangle$. We need to homogenize C_2 as $x^2 - yz = 0$, so the points on C_2 are of the form $\langle x : x^2 : 1 \rangle$ or $\langle 0 : 1 : 0 \rangle$. They share no points at infinity, then, and we are reduced to the case where $x = x^2$, so $x = 1$. Thus, our point of intersection is $\langle 1 : 1 : 1 \rangle$.
- (b) , First, we need to homogenize both equations:

$$\begin{aligned} C_1 : x - y - z &= 0 \\ C_2 : x^2 - y^2 + 2z^2 &= 0. \end{aligned}$$

We then describe the points on C_1 and C_2 in projective coordinates:

$$\begin{aligned} C_1 : & \{ \langle x : x - 1 : 1 \rangle, \langle 1 : 1 : 0 \rangle \}, \\ C_2 : & \{ \langle x : \pm \sqrt{x^2 + 2} : 1 \rangle, \langle 1 : 1 : 0 \rangle \}. \end{aligned}$$

They agree at infinity, then. Back in the affine plane, we solve

$$\begin{aligned}x - 1 &= \pm\sqrt{x^2 + 2} \\x^2 - 2x + 1 &= x^2 + 2 \\2x &= -1 \\x &= -\frac{1}{2},\end{aligned}$$

but plugging $-\frac{1}{2}$ back into x , we get

$$\pm\sqrt{\left(-\frac{1}{2}\right)^2 + 2} = \pm\sqrt{\frac{9}{4}} = \pm\frac{3}{2}$$

and

$$-\frac{1}{2} - 1 = -\frac{3}{2},$$

so we get agreement only when taking the negative square root. Finally, we can characterize all points of intersection as

$$\left\{ \left\langle -\frac{1}{2} : -\frac{3}{2} : 1 \right\rangle, \langle 1 : 1 : 0 \rangle \right\}.$$

□

3. Find a parametrization for all rational points on the curve $y^2 = x^3 - 2x^2 - 4x + 8$. (Hint: There is a point on the curve, a singular point, with the property that any line through that point intersects the curve with multiplicity 2. Because this curve is singular, it has genus 0.)